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Regional Brain Morphometry and Lissencephaly in the Sirenia

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Abstract. Neuroanatomical structure was examined in the brains of West Indian manatees *(Trichechus manatus)* using computer-based morphometric methods. Although manatees have a small relative brain size, volume estimates of the major brain regions indicate that the telencephalon comprises 71% of total brain volume and is 90% cerebral cortex. These values are comparable to those seen among a diversity of taxa having large relative brain size, including many primates. Manatee brains also exhibit well-defined cortical lamination. The measured gyration index (an index of cortical folding) was 1.06, representing a highly lissencephalic condition. These findings demonstrate that small relative brain size and lissencephaly do not constrain the elaboration of internal brain structures. The marked lissencephalic condition is unusual for brains of this absolute size range, and may be related to the thickness of the cortical gray matter and underlying white matter.

Introduction

Recently [O'Shea and Reep, 1990], we considered relative brain size in the Sirenia and its possible relationship to various life history traits. One of the limitations of this approach is that relative brain size reveals little about functional organization. In the absence of experimental physiological data, functional inferences must be made through the study of internal brain structure. Neuroanatomical specializations unique to particular taxa may be visible only on examination of stained brain sections. Such taxon-related characters of the nervous system have been documented for several groups of mammals and provide a basis for constructing phylogenies [Johnson et al., 1982; Kirsch and Johnson, 1983]. Examples include the convoluted olfactory tubercle of armadillos and cortical cell aggregates (barrels) related to mystacial vibrissa function in certain rodents, rabbits and marsupials [Johnson, 1980]. Likewise, in the presumptive sensorimotor cortex of manatees, deeply situated cell aggregates are hypothesized to be related to perioral tactile bristle function [Reep et al., 1989]. Similar cell aggregates are also seen in dugongs *(Dugong dugong)* [Dexler, 1913] and their presence appears to be a uniquely sirenian character.

The consideration of regional brain anatomy represents an intermediate approach between assessment of brain-body weight data and detailed analysis of internal specializations. The relationship between behavioral traits and system-level neuronal operations may be illuminated when all three of these approaches are combined. For example, Johnson et al. [1986, 1987] observed that visual thalamic and brainstem nuclei are reduced in manatee brains, whereas auditory, trigeminal, somatosensory and vagal nuclei are well developed. These findings concur with assessment of the relative importance of these systems in sirenian behavior [Hartman, 1979]. Stephan and Andy [1964] pioneered the use of quantitative regional brain morphometry as a method by which to compare brains across taxa. In this approach, the relative sizes of specific brain regions are compared rather than overall brain volumes. However, the use of regional morphometric data transforms the problem from interpreting differences in overall brain size to inter-

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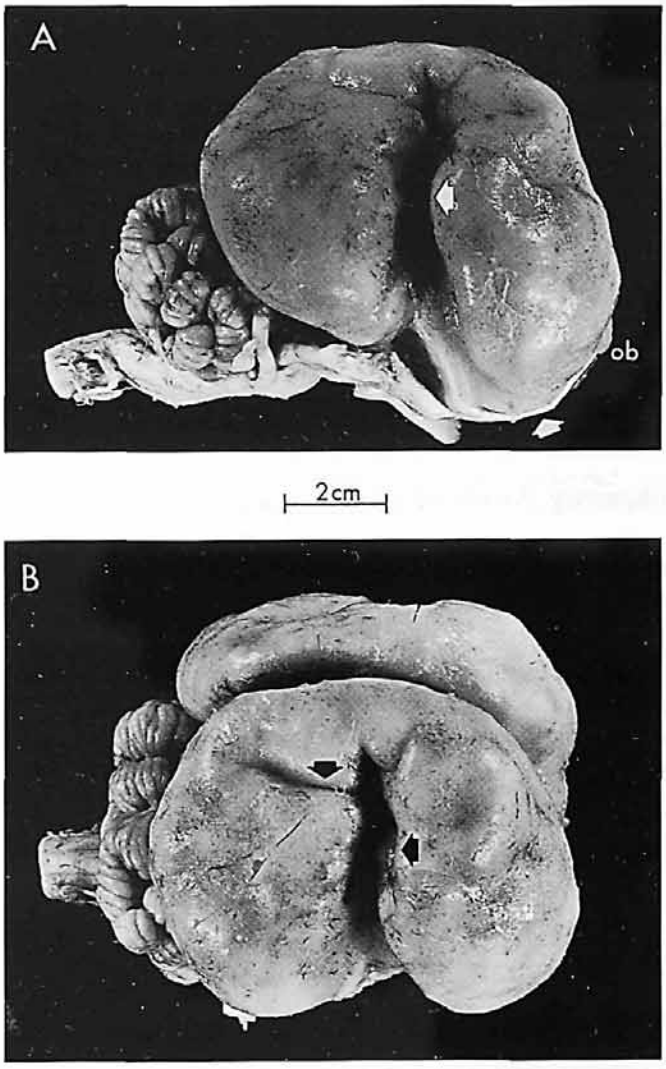


Fig. 1. Gross appearance of an adult *T. manatus* brain (specimen M-387). **A** Right lateral view. Note the vertical orientation of the lateral fissure (horizontal arrow) and ventralward rotation of the hemispheres rostrally. Arrow at bottom right denotes the lateral olfactory tract. ob = Olfactory bulb. **B** Right dorsolateral view. Note the pronounced lissencephaly of the cortical surface aside from the deep lateral fissure (horizontal arrow) and another fissure emanating caudally from it (vertical arrow).

preting differences in the sizes of specific regions. In strict terms, morphometric measurements are directly interpretable only in reference to electrophysiological maps and other experimental data from the species under investigation. Otherwise, such knowledge for other taxa must be used to make inferences for a species (e.g. sirenians) for which such data do not exist.

A notable gross feature of sirenian brains is the general absence of convolutions on the cortical surface. This lissencephalic condition has been typical of the Sirenia throughout their evolutionary history, and has been documented for fossil forms and all Recent species [Elliott Smith, 1902; Edinger, 1933; Friant, 1954]. Although sirenian brains are relatively small [O'Shea and Reep, 1990] and lissencephalic, the telencephalon (particularly the cerebral cortex) appears to be extensive and exhibits a well-defined internal architecture in the manatee *Trichechus manatus* [Reep et al., 1989]. To investigate further the relationships among these features, quantitative morphometric methods were used to examine internal brain organization and the degree of cortical folding in *T. manatus*. Herein, we present these data, compare them with similar data for other mammalian taxa, and discuss the unusual combination of brain traits found in the Sirenia.

Materials and Methods

Four brains of Florida manatees, *T. manatus*, were examined for internal morphometric analysis. All brains were obtained from fresh carcasses (time since death known or estimated to be less than 12 h) collected through a carcass salvage program. In cases M-412 (245-kg female) and M-450 (590-kg male) the brains were extracted, weighed after removal of meninges and immersed in 10% neutral buffered formalin. In cases M-433 (281-kg male) and M-387 (485-kg male; also assigned Wisconsin Comparative Neurobiology Collection No. 84-49) the head was severed from the carcass and perfused in situ by gravity-fed bilateral cannulation of the carotid arteries. Blood was flushed with 10–15 liters of 0.9% phosphate-buffered saline. This was followed by 10–15 liters of 4% phosphate-buffered paraformaldehyde fixative. The brain was then extracted, weighed after removal of meninges and stored in fixative. Three brains (M-412, M-433 and M-450) were cut into coronal slices, 1–2 cm thick, and the caudal face of each slice photographed with calibration. Brain M-387 was embedded in celloidin, sectioned coronally at 40 µm and stained for Nissl substance.

Areal measurements of brain regions were made from the calibrated photographs of specimens M-412, M-433 and M-450, using a digitizing tablet and morphometry software (program PM) developed by Southern Micro (Atlanta, Ga., USA). For analysis of brain M-387, slides were placed on a light table and projected (with calibration) through a video camera equipped with a macro lens. Each image was digitized and analyzed morphometrically using the same software. In all cases the perimeters of measured areas were traced using a hand-held cursor. The left and right sides were measured independently for bilateral structures. Ventricular spaces were not included in the measurements, consistent with the practice of Stephan et al. [1981]. The white matter and overlying gray matter were both included in the areal measurements of cerebral cortex, consistent with the practice of Frahm et al. [1982]. An estimate of relative volume was obtained for each brain region, by summing the areal measurements for that region and calculating the percent

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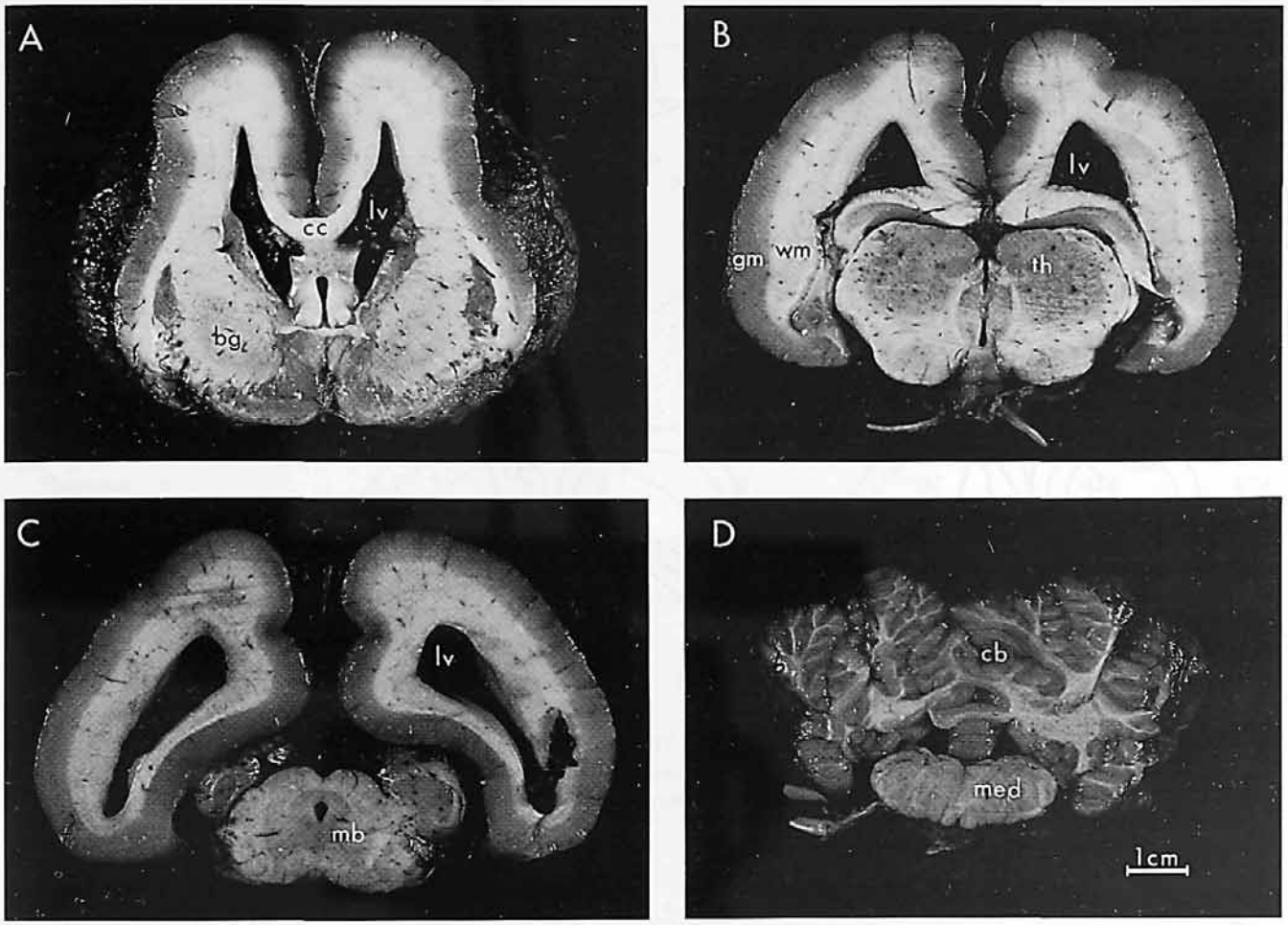


Fig. 2. Photographs of 4 slices from brain M-433, arranged in rostrocaudal order. Note in **ABC** the thick cortical white matter and large lateral ventricles, typical of all specimens examined. bg = Basal ganglia; cb = cerebellum; cc = corpus callosum; gm = cortical gray matter; lv = lateral ventricles; mb = midbrain; med = medulla; th = thalamus; wm = cortical white matter.

contribution to the total of all summed areal measurements for that brain.

Two brains were used to estimate the degree of cortical folding. Twenty-five evenly spaced coronal sections through the cerebral cortex of specimen M-387 were analyzed, representing a sampling interval of 2 mm. The brain from a fifth animal, specimen 84-14 (a 771-kg female, 349-gram brain weight; also assigned Wisconsin No. 84-58), was a perfused specimen embedded in celloidin and sectioned horizontally at 40 μm. Seventeen evenly spaced sections were analyzed from this brain at a sampling interval of 2.9 mm. Using the above morphometric system, separate measurements were made of the exposed cortical perimeter (excluding the olfactory bulb, taenia tecta and hippocampal formation) and the perimeter of the cortex hidden from the surface by fissures [per Zilles et al., 1988a]. Left and right sides were measured independently. The ratio of total to exposed perimeter is the gyration index (GI) of Zilles et al. [1988a]. Brains M-387 and 84-14 were also used to estimate cortical surface area from the perimeter measurements described above, by multiplying the total sampled perimeter by section thickness and the distance between samples, with minor adjustments for excess sections at each end of the sample range.

Slices from specimens M-412, M-433 and M-450 were used to obtain accurate measurements of average cortical thickness. Because these tissues were fixed but not dehydrated or embedded, there was minimal shrinkage. In contrast, manatee brain tissue embedded in celloidin for sectioning exhibits shrinkage of approximately 30% [Reep et al., 1989]. Several spaced measurements of cortical gray matter thickness were made in each slice containing cerebral cortex. Care was taken to sample those regions where the cortex appeared to be sliced perpendicular to its radial axis, exhibited no pronounced curvature, and displayed a defined boundary between the cortical gray matter and underlying white matter. These procedures yielded 40–50 measurements per brain.

Brain volumes were determined from the sample of 13 animals described by O'Shea and Reep [1990]. Average fresh brain weight (364 g) was divided by 1.036 g/cm^3, the average specific gravity of fresh brains [Frahm et al., 1982].

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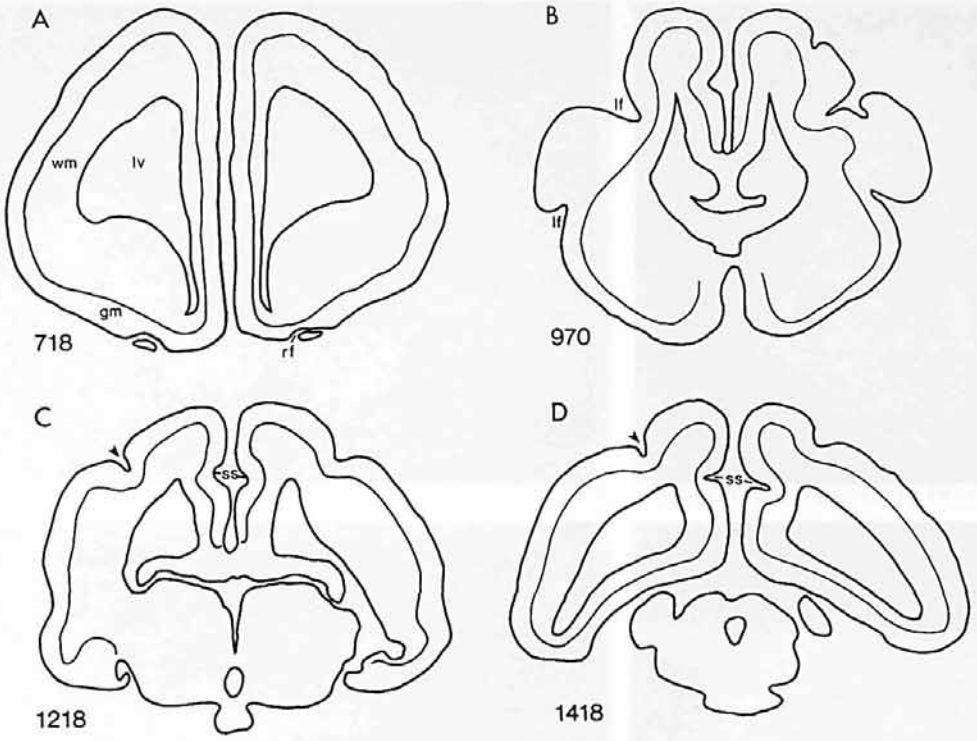


Fig. 3. Four line drawings of coronal sections from brain M- 387, illustrating the prominent fissures and sulci. The sections are in rostrocaudal sequence and numbered. Note the large lateral ventricles. gm = Gray matter of cerebral cortex; lf = lateral fissure; lv = lateral ventricle; rf = rhinal fissure; ss = splenial sulcus; wm = white matter of cerebral cortex. Arrowhead denotes an unnamed sulcus on the dorsolateral surface, which runs caudally from the lateral fissure.

Results

Internal Morphometrics

Notable gross features typical of all manatee brains examined included the generally lissencephalic cortical surface, ventral rotation of the hemispheres rostrally, a deep and vertically oriented lateral fissure, and a fissure emanating caudally from the dorsal margin of the lateral fissure (fig. 1). Even in slices without staining there is good contrast between gray matter (fig. 2). Thus, the boundaries between the major brain regions are well defined, as is the thickness of the cortical gray matter. The lateral ventricles are large (fig. 2, 3).

The telencephalon averages 71% of total brain volume, with cerebral cortex representing about 90% of this brain region (table 1). There were no consistent left-right differences in estimated cerebral cortical volumes. In cases M-412 and M-433, the estimated percent volume was greater for the right hemisphere, whereas in cases M-450 and M-387 it was greater for the left. However, in all cases the left-right difference was less than 6%, probably within the range of sampling error.

Determinations of regional percent volumes were made on relatively few slices in specimens M-412, M-433 and M-450, compared to the number of sections used from brain M-387, which was processed histologically. To estimate the precision of the data obtained from the 3 sliced specimens, we calculated the correspondence between estimates of percent volume by using different numbers of slices from brains M-412 and M-450. Two comparisons were made for each of these brains: measurements made from one half of the slices were compared with those made from the other half, and measurements made from half the slices were compared to those made from all the slices (table 2). In each case, for each brain region measured, a percent correspondence factor was determined by dividing the difference in percent volume estimates by the average of the two estimates. There was more than 90% agreement in the volume estimates of cerebral cortex, basal ganglia and telencephalon (table 2). This was true when two half-samples of

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Table 1. Percent volume estimates of the major brain regions in 4 Florida manatees

Parameter	Specimen				Mean
	M-412	M-433	M-450	M-387	
Brain weight, g	392	346	455	385	
Sections, n	17	7	15	49	
Telencephalon					
Cortex	67.25	58.93	64.44	64.74	63.83
Basal ganglia/septum	7.23	7.06	5.78	6.81	6.72
Amygdala	0.69	0.80	1.13	0.20	0.71
Total	75.17	66.79	71.35	71.75	71.26
(Cerebral cortex as percent telencephalon)	(89.46)	(88.23)	(90.31)	(90.23)	(89.57)
Diencephalon					
Thalamus	4.55	4.79	5.58	4.05	4.74
Hypothalamus	0.72	1.76	0.58	0.43	0.87
Total	5.27	6.55	6.16	4.48	5.61
Mesencephalon	3.32	3.23	4.17	2.51	3.31
Pons	1.95	3.62	4.44	2.91	3.23
Cerebellum	11.25	16.50	12.00	15.79	13.85
Medulla	2.92	3.28	1.81	2.48	2.62

The brain weight and number of sections measured are given for each specimen.

7–9 sections each were compared within the same brain, when estimates obtained from half the slices were compared to those made from all the slices, and when estimates were compared between the 2 brains. Undoubtedly, this reflects an adequate frequency of sampling of these rather large brain regions and suggests that relatively few slices can yield a reasonable estimate of percent volume for larger brain regions. Volume estimates of smaller brain regions were highly variable, with low percent correspondence within and between brains. For these regions, volume estimates from brain M-387 (table 1), with 49 sections, are probably the most reliable.

Lissencephaly and the GI

In lateral view, sirenian brains are strikingly lissencephalic except for the lateral fissure and another caudally directed sulcus (fig. 1). Other shorter and shallower sulci are more variable among specimens. The average measured GI in our material was 1.06 (table 3), with no left-right difference in either case. The principal sulci that produce a hidden cortex include the rhinal fissure, lateral fissure, splenial sulcus and an unnamed sulcus on the dorsolateral surface. The cortical surface area averaged 127 cm^2 (table 3). The surface area is probably underestimated because of shrinkage inherent in the celloidin embedding procedures used in these 2 cases.

Cerebral cortical thickness and brain volume pertain to considerations of lissencephaly and gyration as discussed below. The cortical thickness averaged 4 mm (table 4) and the average brain volume was 351 cm^3.

Discussion

Comparisons of the contributions of the major brain regions to overall brain volume (fig. 4) indicate that manatees exhibit a relative telencephalic volume comparable to primates and markedly greater than for bats and insectivores (corresponding data for un-

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Table 2. Correspondence among percent volume estimates in 2 Florida manatee brains

Slices, n	Specimen			
	M-412		M-450	
	8+9	9+17	7+8	8+15
Telencephalon				
Cortex	93	97	97	98
Basal ganglia/septum	96	99	93	96
Amygdala	47	79	60	83
Total	94	97	98	99
Diencephalon				
Thalamus	82	91	94	97
Hypothalamus	45	79	87	94
Total	91	96	94	97
Mesencephalon	94	97	31	60
Pons	30	61	75	88
Cerebellum	55	82	67	86
Medulla	61	84	42	76

The values shown are the percent correspondence between measurements made from different numbers of slices. The total number of slices was 17 for brain M-412, 15 for M-450.

Table 3. Cortical data derived from spaced sections in 2 Florida manatee brains

Brain	GI	Cortical surface area, cm^2		
		left	right	total
M-387 (coronal)	1.05	56	60	116
84-14 (horizontal)	1.07	68	69	137
Mean	1.06	62	64	127

Table 4. Cortical gray matter thickness measured directly from slices of brains from 3 Florida manatees

Brain	Cortical thickness, mm	
	(mean)	SD
M-412 (n = 48)	3.99	0.33
M-433 (n = 45)	4.10	0.52
M-450 (n = 51)	3.98	0.37
Mean	4.23	0.41

gulates, carnivores and other groups have not been reported). This characteristic appears to be representative of Recent sirenians: the telencephalon of dugongs comprises 73% of total brain volume [Pirlot and Kamiya, 1985; data from 1 brain specimen], similar to the 71% derived for manatees in this study.

Alternatively, we can compare the relative size of cerebral cortex, expressed as percent telencephalic volume or as percent total brain volume. The values for manatees derived from the present data (90 and 63%, respectively) are comparable to those of 93 and 68% for dugongs, based on data from the single specimen of Pirlot and Kamiya [1985]. The proportion of brain volume devoted to cerebral cortex in sirenias is greater than for taxa grouped as insectivores, and comparable to primates (fig. 4). Frahm et al. [1982] focused on the neocortex in their comparative study. However, because this approach excludes olfactory cortical areas it is biased against taxa in which olfaction is a dominant sensory modality. As discussed by Reep [1984, 1988], this tends to perpetuate the long-standing anthropocentric view of olfaction as a 'primitive' sensory modality, although there is no evidence to support this attitude as it relates to mammalian phylogeny. Therefore, all cortical areas are included in our calculations of the cerebral cortex as percent telencephalon (fig. 4).

These results suggest that the sirenian cerebral cortex is elaborated to a degree not evidenced by the low encephalization quotients of this group [see O'Shea and Reep, 1990], if one accepts the conventional argument that larger relative brain sizes result from elaboration of telencephalic components [Hofman, 1982]. However, it appears more appropriate to view the low encephalization quotient in manatees as representative of large relative body size rather than small relative brain size [O'Shea and Reep, 1990; see also Deacon, 1990]. Further support for this point of view is the finding that the internal cytoarchitecture of manatee cerebral cortex is well defined, including such features as multiple distinctly laminated areas in the frontal lobe, pronounced radial and tangential striations, and specialized clusters of cells in layer VI of the presumptive sensorimotor cortex [Reep et al., 1989]. These anatomic data are evidence that the manatee cerebral cortex is well developed and capable of an extensive amount of complex information processing.

The average measured GI of 1.06 places manatee brains among the most lissencephalic, and fossil evi-

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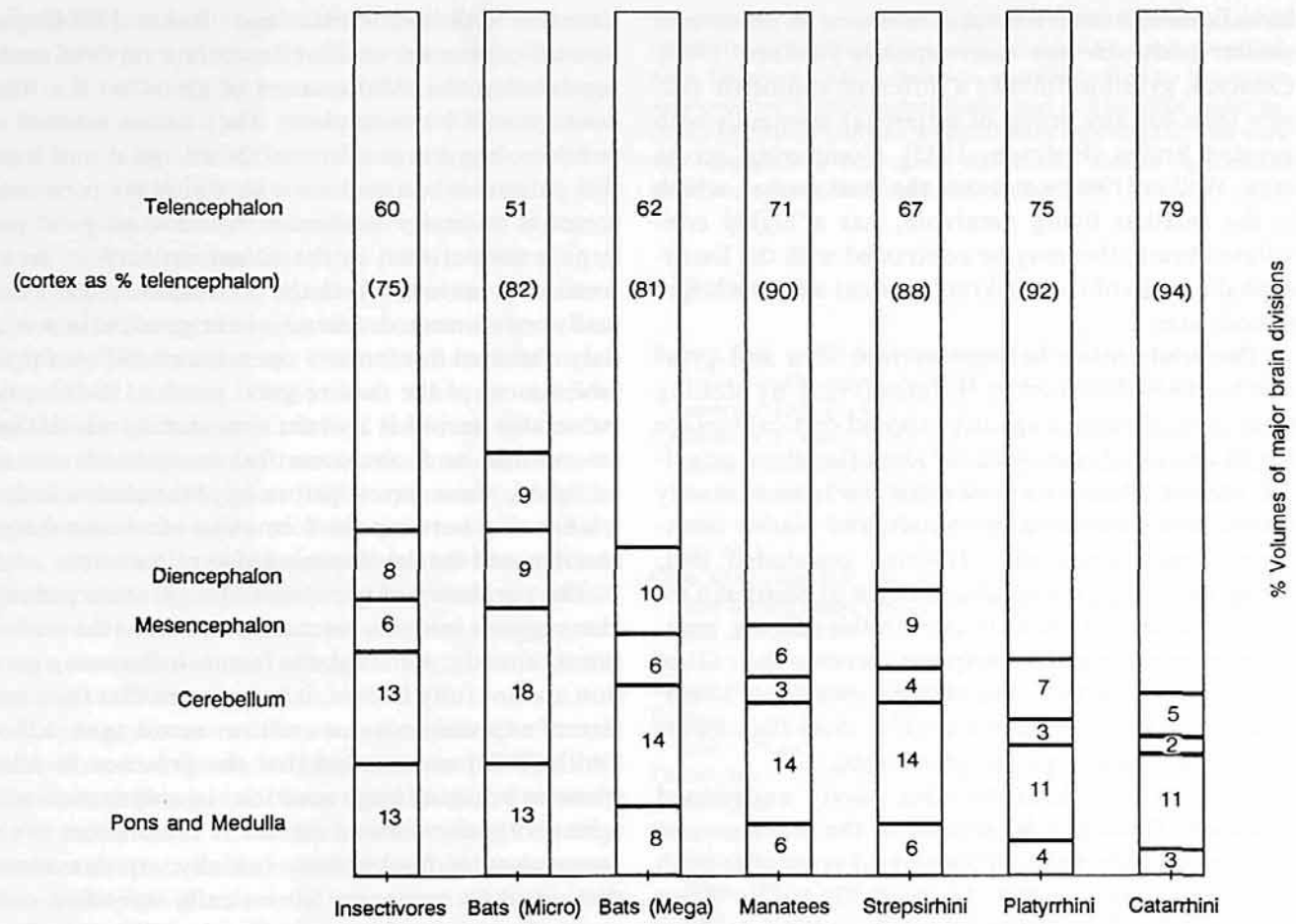


Fig. 4. Relative volumes of the major brain divisions in several groups of mammals. The data on manatees are from the present study. The data on Megachiroptera (7 species) are from Stephan et al. [1974], those on Microchiroptera (15 species) from Stephan and Pirlot [1970], and these on insectivores (28 species), strepsirhine (20 species), platyrrhine (13 species) and catarrhine (13 species) primates from Stephan et al. [1981]. The numbers in parentheses represent the total cerebral cortex as percent telencephalic volume; values for taxa other than manatees were derived from the above references and Frahm et al. [1982] and are overestimates because they include the nucleus accumbens and the amygdala. Note that manatees have a well-developed telencephalon with a cerebral cortex comparable to that of primates and larger than that of insectivores and bats.

dence suggests that lissencephaly has been a constant feature among the Sirenia [Edinger, 1933, 1939]. Highly lissencephalic brains have GIs near 1.0, whereas highly convoluted brains such as those of some cetaceans have GIs greater than 4.0. Ungulates generally exhibit intermediate values, from 2.0 to 2.8 [Hofman, 1985]. The lissencephalic condition in manatees is striking because of the large absolute size of the brain; all other mammalian brains weighing over 300 g are gyrated. These include species among the orders Artiodactyla, Carnivora, Perissodactyla, Pinnipedia, Primates and Proboscidea [Crile and Quiring, 1940; Hofman, 1985; Martin and Harvey, 1985]. The general trend for large brains to be gyrated led Jerison [1983] to propose that absolute size is the determining factor in this process across mammalian taxa. However, it now appears that this is an overgeneralization, and that the pattern and degree of cortical folding are group-specific. In primates the degree of fissuration is species-typical and correlates well with absolute brain size [Hershkovitz, 1970; Jerison, 1983]. However, taxa grouped as prosimians show less dependence of GI on absolute brain size than those grouped as anthropoids [Zilles et al., 1988b]. Within rodents, capybaras

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have fissured brains whereas the brains of beavers of similar body size are lissencephalic [Welker, 1990]. Cetacean gyration follows a different allometric pattern than for any group of terrestrial mammals with gyrated brains [Hofman, 1985]. Comparing across taxa, Welker [1990] notes that the least weasel, which is the smallest living carnivore, has a highly convoluted brain; this may be contrasted with the lissencephalic brain of the muskrat, a rodent which is larger in body size.

The relationship between surface area and gyration has been discussed by Hofman [1985]. By plotting total cortical surface against exposed cortical surface for 44 species of mammals, he identified three principal groups: those with lissencephalic brains (mostly insectivores); terrestrial mammals, and marine mammals (mostly cetaceans). Hofman concluded that, across mammals, convolutions begin to occur at a total surface area of about 10 cm^2. In this scheme, manatees represent a major exception, because their GI of 1.06 and total surface area of approximately 130 cm^2 extends the lissencephalic condition more than an order of magnitude larger in surface area.

Gyration and fissuration are poorly understood processes. Although the sutures of the manatee cranial vault remain unfused for several years after birth [O'Shea and Reep, 1990], this probably has no direct influence on the degree of adult cortical folding. Growth of bones of the cranial vault is directed by the underlying brain tissue, not vice versa [De Beer, 1937]. Also, gyral formation does not appear to be a simple mechanical response to skull constraints, because experimental partial removal of skull and brain tissue does not alter the basic fissuration pattern [Barron, 1950].

The formation of cortical gyri occurs only after the generation of cortical neurons [Richman et al., 1975; Goldman-Rakic and Rakic, 1984]. Thus, regional variations in cell proliferation rate appear to have no immediate effect on normal gyration. Disruptions of the proliferative phase of neurogenesis are known to influence subsequent brain development, including gyral patterns [Richman et al., 1975; Caviness and Williams, 1984]. However, later occurring events such as cell death and the establishment of neuronal connections are likely to be involved in such cases. Experimental findings in fetal rhesus monkeys suggest that formation of neuronal connections and associated competition for synaptic space may be significant factors in the formation of normal and abnormal gyral patterns. Goldman-Rakic and Rakic [1984] performed prefrontal cortical resections on fetal monkeys during the third quarter of gestation (i.e. after neurogenesis was complete). They found marked alterations in gyral patterns in the occipital and superior parietal lobes, regions with which the prefrontal cortex is normally connected. Anomalous gyral patterns were not seen in the somatosensory or motor cortices, areas with which the prefrontal cortex is normally not connected. Alterations in gyral pattern were only obtained in monkeys operated on before the establishment of the mature gyral pattern. Because the vulnerable period is also the time during which thalamocortical and corticocortical connections are established, these investigators hypothesized a causal relationship between the formation of neuronal connections and the development of gyral patterns.

The regularity of group-specific gyration patterns also suggests intrinsic mechanisms such as those mentioned already. Although the factors influencing gyration are not fully known, it is apparent that their pattern of expression is not uniform across taxa. Elliott Smith [1902] commented that the presence in adult manatee brains of large ventricles in conjunction with a lissencephalic cortical surface is reminiscent of the appearance of fetal brains. Initially, we considered that, if there were an intrinsically specified time course for the development of convolutions, perhaps the Sirenia exhibit a retardation of this process. Such an explanation would be consistent with other sirenian life history traits and small neonatal brain size [O'Shea and Reep, 1990]. However, because cortical cytoarchitecture and lamination are well defined in manatees, any such retardation has had no apparent effect on neurogenesis, migration and the establishment of neuronal connections. Thus, if intrinsic mechanisms such as establishment of connections and competition for synaptic space also operate in manatees, retardation of the growth of fetal brain would not explain how the GI of manatees could be such an outlier among mammals with brains in this absolute size range.

The thickness of the cortical gray matter and underlying white matter may play a key role in manatee lissencephaly. Our measured value of 4 mm for average thickness of the cortical gray matter in manatees is much greater than for most mammals investigated, comparing the data for 7 taxa cited by Prothero and Sundsten [1984]. However, these thickness data are taken from histologically processed specimens, which

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introduces substantial amounts of shrinkage and variations in degree. Generally this is not controlled for, but Rockel et al. [1980] reported 18% linear shrinkage in *Macaca mulatta* and we estimated 35% in manatees [Reep et al., 1989]. Even if a shrinkage-corrected value of 2.6 mm is used for manatees, this value still lies above the overall mammalian regression line generated by Prothero and Sundsten [1984]. According to mechanical models of gyral formation, increased thickness of the cortical gray matter contributes to lissencephaly [Richman et al., 1975; Prothero and Sundsten, 1984; Hofman, 1985]. Such schemes consider that the establishment of cortical thickness occurs before the development of convolutions and influences their formation. Hofman [1985] appears to view the hypothesized causality of these processes in the reverse, such that in lissencephalic brains cortical volume increases through combined increases in cortical surface area and thickness; he suggests that gyrated brains increase volume almost entirely through expanding surface area.

A striking feature of the manatee cerebral cortex is the thickness of the underlying white matter (fig. 2A–C), although this has not been quantified in our study or for other taxa. If, as Richman et al. [1975] suggested, a deep stratum can influence the development of convolutions in an overlying stratum, the apparent thickness of the white matter in manatees may constrain the gray matter to a lissencephalic condition. The plausibility of this explanation depends on the relative rates of growth in thickness of the gray matter and white matter strata, which are unknown for the Sirenia.

Regardless of the ultimate explanation of lissencephaly in manatees, the relative independence of neuronal proliferation and fissuration might explain why manatees are lissencephalic but exhibit a well-defined cortical lamination.

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